



The Making and Breaking of Radioactive Molecules

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Radioactive Molecules at NEPTUNE

- The Standard Model fails to describe 95% of the mass and energy in the Universe, matter-antimatter imbalance, ...
- Our group is searching for violations of fundamental symmetries in radioactive molecules [1]
- Trapping (radioactive) molecular ions at NEPTUNE in a Penning ion trap can amplify electroweak nuclear properties (e.g., the nuclear anapole moment) by more than **12 orders of magnitude** [2]

$$W_{PV} \sim \frac{< - |H_{PV}| + >}{E_- - E_+}$$

Atom: ~1 eV
Molecule: ~10 μ eV
Molecule + B: ~1 peV

Factor >10¹² amplification!

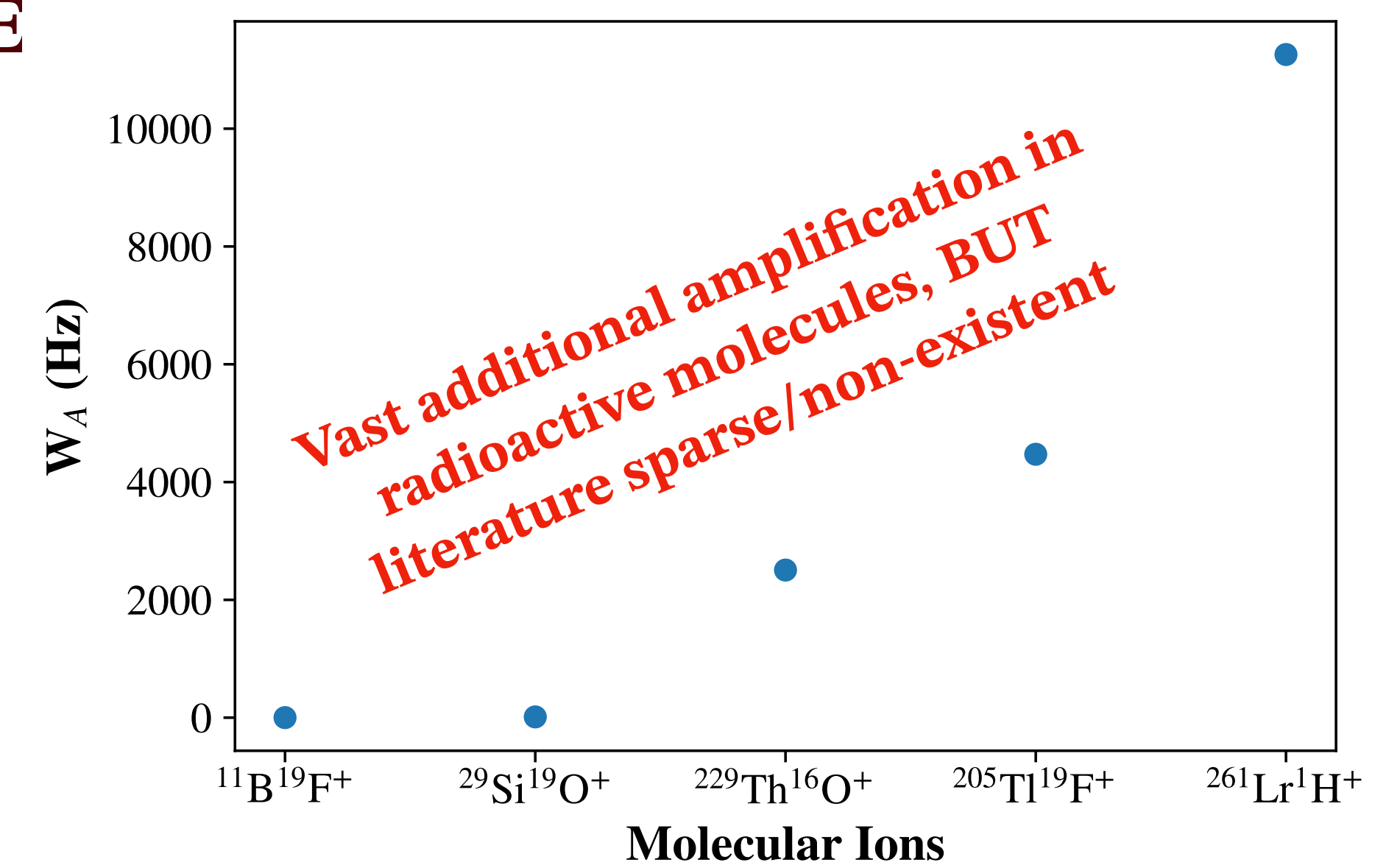


Fig 1. Diatomic molecular ions with sizable weak matrix elements W_A [1]

Piezo-based Laser Ablation Ion Source

- Stable and long-lived molecules** in vacuum can be created using laser ablation of solid targets
- Ion acceleration with noble gases and rapid expansion into vacuum creates "**cold**" **molecules** in the lowest ro-vibrational states
- Current designs often use solenoid gas valves, leading to long (>100 μ s) gas pulses and, thus, high gas loads that reduce cooling efficiency
- Piezoelectric valves [3] allow for **ultra-short gas pulses** (~10 μ s)
- Design and commissioning of the ion source in collaboration with Prof. Yan Zhou's at UNLV

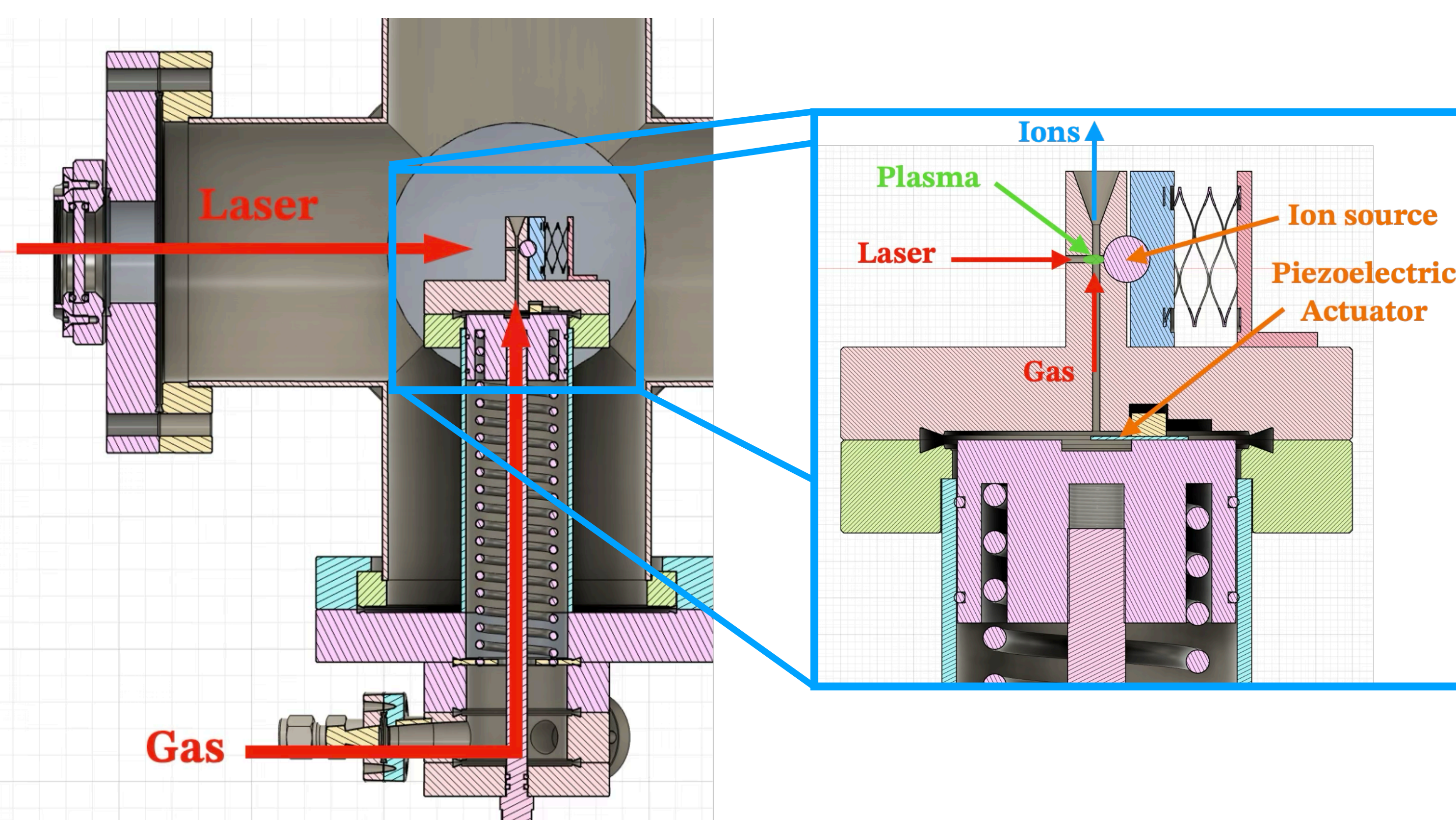


Fig 2. A three-dimensional cross section of the Piezoelectric Laser ablation ion source

The Si₃N₄ Dissociation Window

- 2 mm × 2 mm** Si₃N₄ window, just **~21 nm thick** ($\approx$ 70 atomic layers)
- Impinging molecular beam get ro-vibrationally excited through collisions far above the molecular **dissociation** threshold
- Withstands pressure differentials up to 200 mbar $\rightarrow$ allows for gas-filled RFQ trapping section after dissociation for efficient cooling of **isotopes of interest** OR the creation of **molecules of interest**



Fig 5. The Si₃N₄ window installed on a CF flange [6].

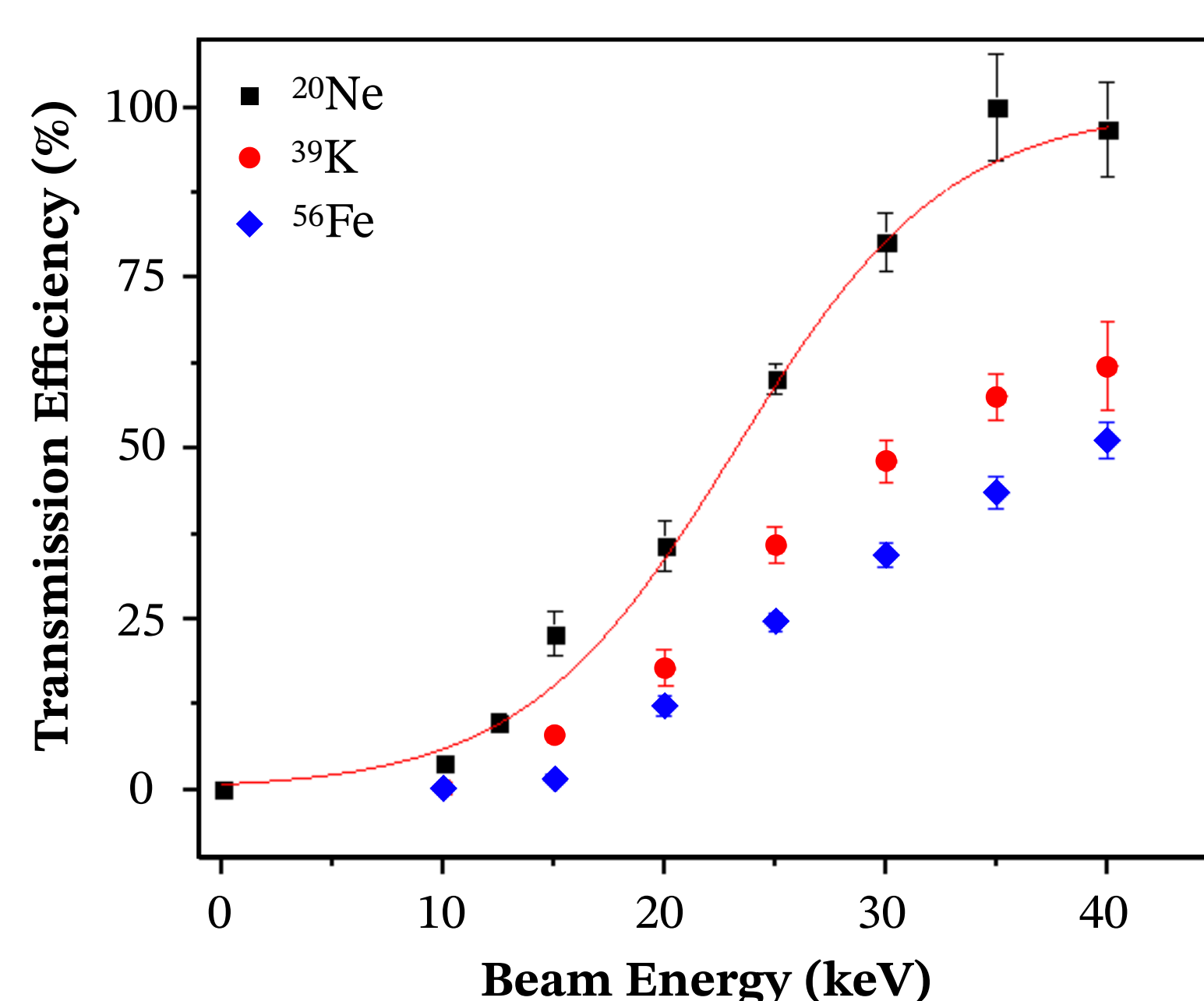


Fig 6. First transmission efficiency results of the CID-GC [6].

Collision-Induced Dissociation Gas Cell (CID-GC)

- Short-lived molecular ions require radioactive beam facilities
- Beam facilities do not typically provide the molecules of interest, while molecular contaminants and isotopes of interest trapped in molecules pose an increasing challenge to stopped beam experiments

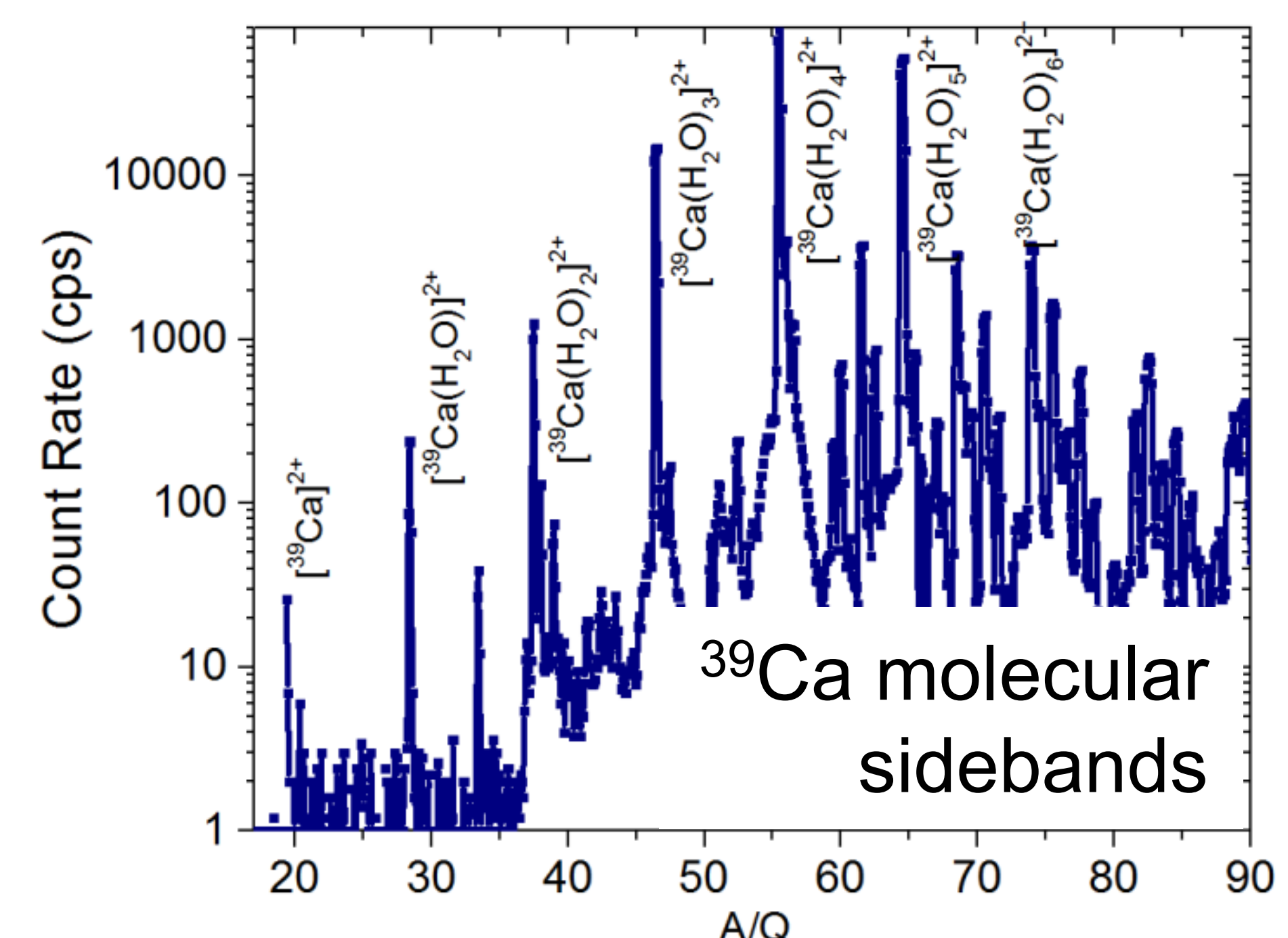


Fig 3. Example of a molecular contaminant spectrum (Courtesy to R. Ringle).

- Various groups (e.g. at GSI [4] and TRIUMF [5]) have applied CID in RFQ cooler/bunchers with varying degrees of success and efficiency, while laser dissociation is molecule specific and requires extensive knowledge of the molecule

$\rightarrow$ With the LEBIT group at FRIB, we are developing a general-purpose device to **break up molecules** through collision-induced dissociation [6]

$\rightarrow$ Molecules pass through an ultra-thin Si₃N₄ membrane to induce dissociation, followed by RFQ for trapping and/or molecule formation

$\rightarrow$ Molecular spectroscopy performed at Texas A&M in new general-purpose action spectrometer (*see poster of Nick Schnoor*)

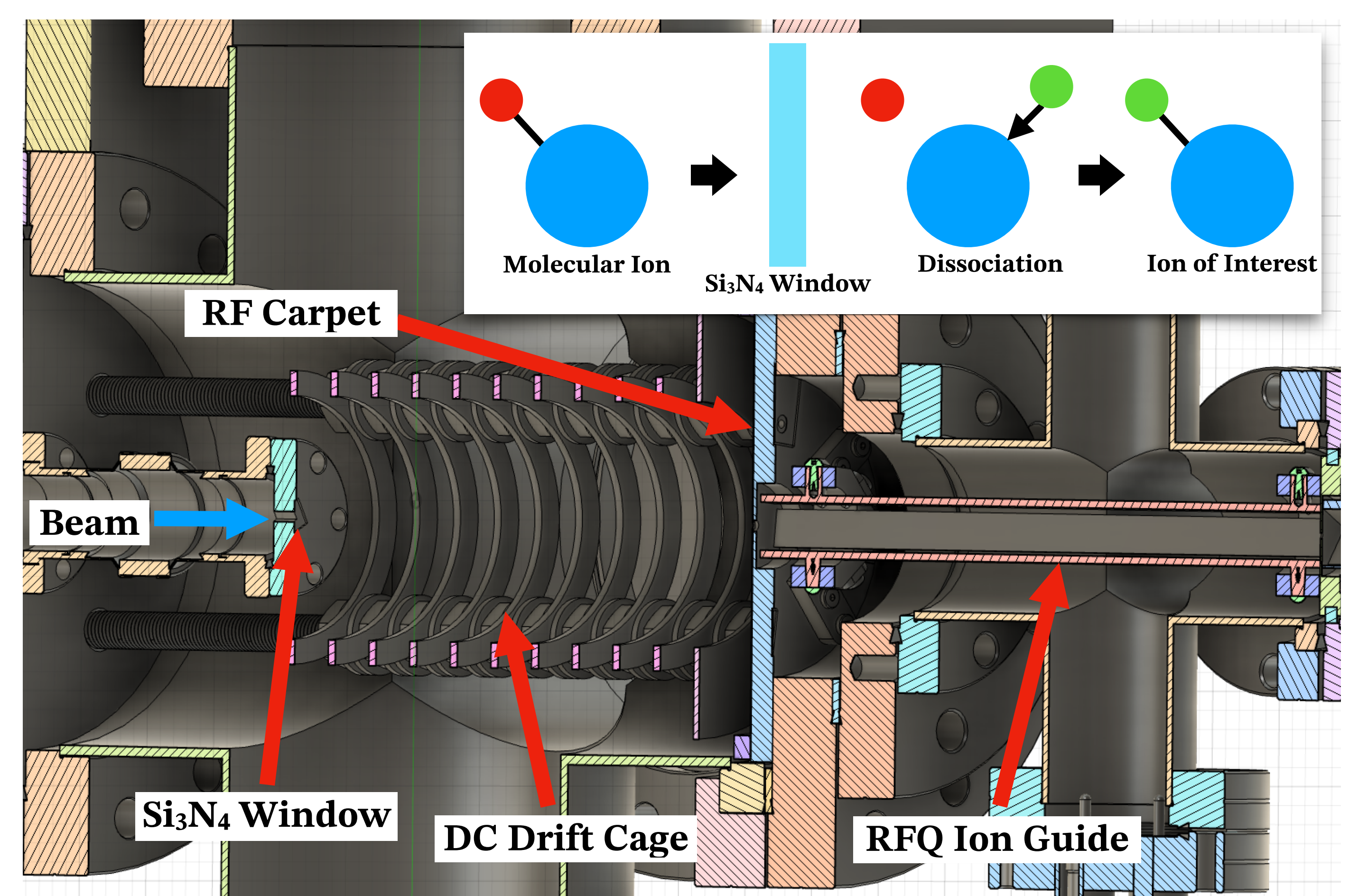


Fig 4. Cross-section of the CID-GC [6] and illustration of the dissociation + formation mechanism (inset).

References

- [1] Arrowsmith et al. *Rep. Prog. Phys.*, **87**, 084301 (2024)
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[3] Catanese A et al. *Rev. Sci. Instrum.*, **89**, 103115 (2018).
[4] Greiner F et al. *NIM-B*, **463**, 324-326 (2020).
[5] Jacobs A et al. *Int. J. Mass Spec.*, **482**, 116931 (2022).
[6] Ringle R. Final report: DE-SC0021423 (2024).

Acknowledgment



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